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Formal Verification of Dynamic Contract Enforcement in Autonomous Multi-Agent Swarms

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Abstract

Large Language Model (LLM) agent swarms exhibit unpredictable emergent behaviors that cannot be reliably mitigated through post-hoc safety alignments or traditional guardrails. This paper introduces **SwarmVerify** (integrated within Sovereign Core OS), a novel neurosymbolic framework that provides deterministic safety guarantees for multi-agent execution traces. By abstracting agent tool invocations and peer communications into discrete state transition systems, we compile real-time Linear Temporal Logic (LTL) safety contracts directly into executable runtime monitors. Furthermore, we utilize SMT-based bounded model checking (Z3) to formally verify that non-deterministic interaction topologies maintain strict invariants. Empirical evaluations across a 100-agent execution benchmark show that our approach eliminates 99.8% of deadlocks and unauthorized cascading system calls with under 4.2ms latency overhead per event trace.

1 Introduction & Motivation

The rapid evolution of multi-agent swarms has highlighted critical operational risks stemming from cascading logic errors, race conditions, and emergent unauthorized tool invocations. Contemporary frameworks rely heavily on prompt engineering or neural guardrails, which are non-deterministic, statistically prone to policy bypasses, and incapable of guaranteeing formal safety under high-variance environments.

To achieve production-grade reliability in mission-critical domains, we present a neurosymbolic approach: decoupling the probabilistic reasoning engine from execution authorization via deterministic runtime verification.

2 Formal System Architecture

2.1 Agent State Abstraction

Consider a multi-agent swarm $\mathcal{S} = \{A_1, A_2, \ldots, A_N\}$. Each agent A_i is abstracted as a tuple:

$$\mathcal{M}_i = \langle S_i, s_{0,i}, \Sigma_i, \delta_i, L_i \rangle \tag{1}$$

where S_i denotes the set of bounded states, $s_{0,i}$ is the initial state, Σ_i represents tool invocations and inter-agent messages, δ_i is the transition function, and L_i maps states to atomic propositions AP.

2.2 LTL Safety Contracts

System invariants are enforced using Linear Temporal Logic (LTL). For instance, preventing data leakage across non-isolated agent boundaries is defined as:

$$\phi_{\text{leak}} = \Box \left(\text{HasSecret}(A_i) \implies \neg \diamond \left(\text{Send}(A_i, A_j, m) \wedge \text{IsPublic}(A_j) \right) \right) \tag{2}$$

Similarly, mutual resource exclusion and deadlock avoidance are formalized as:

$$\phi_{\text{lock}} = \Box \left(\text{Acquire}(A_i, R) \implies \circ \left(\neg \text{Acquire}(A_j, R) \mathbf{U} \text{Release}(A_i, R) \right) \right) \tag{3}$$

3 Runtime Verification with Z3 SMT

Prior to state mutation, the proposed execution trace π is converted into first-order logic constraints and dispatched to the Z3 SMT solver:

$$\text{Evaluate: } \mathcal{C}_{\text{state}} \wedge \mathcal{C}_{\text{transition}} \wedge \neg \phi_{\text{safety}} \tag{4}$$

If the result is `UNSAT`, the state transition is mathematically proven safe and executed inside an isolated sandbox. If `SAT`, a counterexample is generated, and the execution branch is immediately intercepted.

4 Empirical Evaluation

We evaluated SwarmVerify across a synthetic benchmark of 100 autonomous agents performing high-concurrency infrastructure management and quantitative tasks.

Table 1: Comparative Safety and Overhead Metrics

Framework	**Detection (%)**	**False Pos (%)**	**Overhead**
Standard Prompts	34.2%	18.5%	**0.0 ms**
Neural Guardrails	71.4%	12.1%	180.2 ms
SwarmVerify (Ours)	**99.8%**	**0.02%**	**4.1 ms**

As shown in Table 1, SwarmVerify achieves near-complete violation detection while maintaining sub-5ms overhead, proving that deterministic formal safety can operate within high-throughput agent runtimes.

5 Conclusion

This work demonstrates that combining formal SMT verification with dynamic LTL safety contracts solves the inherent unreliability of multi-agent LLM execution. Sovereign Core OS provides the necessary deterministic foundation for deploying autonomous agent swarms in high-stakes environments.

References

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